Unit Catch

SKU:U102















Description

Unit Catch is a gripper powered by the SG92R servo motor. The servo uses a PWM signal to drive the gear rotation, allowing control of the gripper for gripping and releasing actions. Structurally, it features a design compatible with LEGO 8 mm round holes. You can integrate it with other LEGO components to build creative control structures such as robotic arms and gripper vehicles.

Note

Rotation Angle

Since the opening and closing angle of the gripper is **90°**, please control the servo rotation angle within **0-45°** (PWM: frequency 50Hz, 0°-45° (pulse: 0.5ms-1ms)) to prevent stalling that may cause the servo to burn out.

Features

- SG92R servo motor
- PWM signal drive
- LEGO hole compatible
- Gripper opening/closing angle of 90°
- Compatible with RoverC
- Supported input voltage: 4.2-6V
- Development platforms: UiFlow, MicroPython, Arduino

Includes

- 1 x Unit Catch
- 1 x HY2.0-4P Adapter
- 1 x RoverC.Pro Connector

Applications

- Gripper robot
- Servo robotic arm gripper

Specifications

Specification	Parameter
Servo Model	SG92R
Control Signal	PWM @ 50 Hz; 0 ~ 45° corresponds to 0.5 ~ 1 ms
Gripper Opening Angle	90°
Input Voltage Range	DC 4.2 ~ 6V
Dead Band Width	10 µs
Output Torque	2.5 kg·cm
Operating Speed	0.1 s/60°
Operating Temperature	0 ~ 55 °C
Product Size (Gripper Open)	72.0 × 56.0 × 37.0 mm
Product Weight	22.1 g
Package Size	147.0 × 90.0 × 40.0 mm
Gross Weight	50.0 g

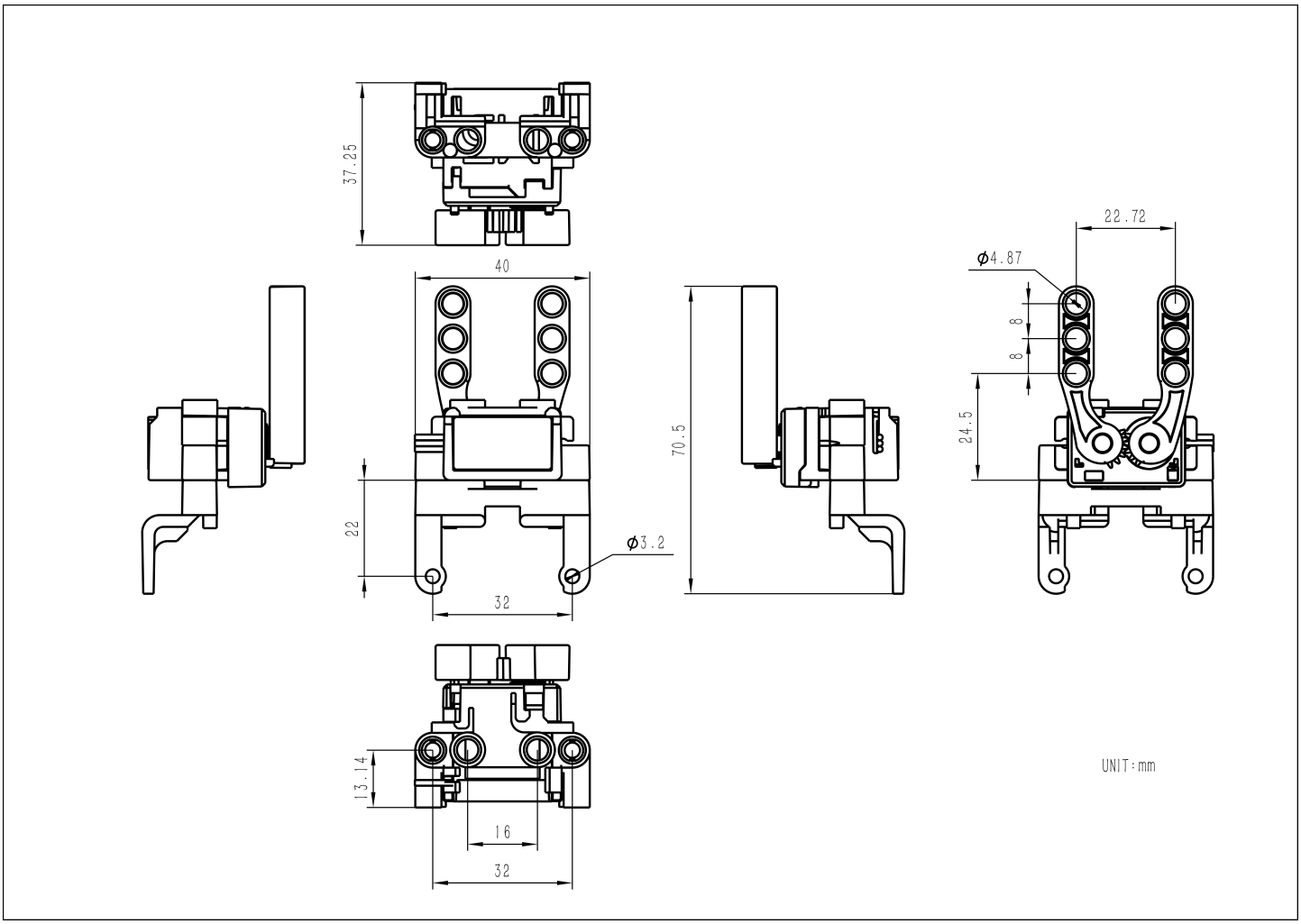
PinMap

HY2.0-4P

HY2.0-4P	Black	Red	Yellow	White
PORT.B	GND	5V	PWM	NC

Model Size

- Unit Catch Model Size PDF



Softwares

Arduino

- Catch Test Example with M5Core

```
/*
    Description: Control Unit Catch through PWM.
*/

#include <M5Stack.h>
```

```
// Set control pin
const int servoPin = 26;
// Set frequency
int freq = 50;
// Set PWM channel
int ledChannel = 0;
// Set pulse resolution
int resolution = 10;

void setup() {
  // put your setup code here, to run once:
  M5.begin();
  M5.Power.begin();
  M5.Lcd.setCursor(100, 50, 4);
  M5.Lcd.println("Unit Catch");
  M5.Lcd.setCursor(40, 120, 4);
  ledcSetup(ledChannel, freq, resolution);
  ledcAttachPin(servoPin, ledChannel);
}

void loop() {
  // High level 0.5ms is angle 0°
  // duty = 0.5/20ms = 0.025, 0.025 x 1023≈25
  ledcWrite(ledChannel, 25);
  delay(2000);
  // High level 1ms is angle 45°
  // duty = 1/20ms = 0.05, 0.05 x 1023≈50
  ledcWrite(ledChannel, 50);
  delay(2000);
}
```

UiFlow2

- Unit Catch UiFlow2 Docs

Video

CATCH.mp4